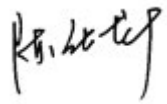
产品规格书

PRODUCT SPECIFICATION

客户名称 Buyer Name	
客户料号 Buyer Part No.	
客户承认签章 Buyers Approval & Signatures	

文件编号 Spec No.		版本	A/0
品名描述 Product Description	LRA 扁平振动马达 LRA Coin Vibration Motor		
型号Part No.	VG0415001W-240H		
送样日期Date			
设计Designed by	审核Checked by	批准Approved by	
陈满			
2025.02.21	2025.02.21	2025.02.21	

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Revised record

Rev. No.	Rev. Date	Page No.	Revised Item	Reason
A/0	2025.02.21	/	Preliminary spec	

1. Applications

This specification provided by Vybronics is applied to model VG0415001W-240H AC linear resonant actuator, which is used for Smart Rings or any device under 10 grams.

2. Storage, Operating Temperature/Humidity Conditions

No.	Item	Condition
2-1	Operating Temperature Range	-20 ℃ ~ +70 ℃
2-2	Storage Temperature Range	-40 ℃ ~ +85 ℃

3. Measurement Conditions, Input Voltage

No.	Item	Condition
3-1	Temperature	20 ± 5 ℃
3-2	Humidity	65 ± 20%RH
3-3	Rated Input Voltage	0.4 ± 0.02 Vrms AC, Sinewave
3-4	Input Voltage Range	0.1 ~ 0.42 Vrms AC
3-5	Operating Attitude	Refer to Figure

Refer to figure：

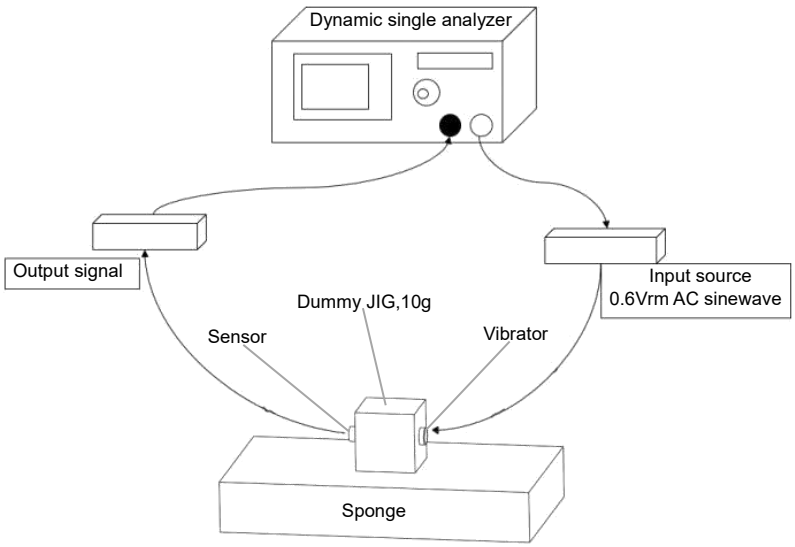


Figure1：An example of measurement method of linear vibrator

Note;

- Dummy Jig size is 20mm x 20mm x 20mm, 10 g (with Sensor).
- Jig should be put it on the sponge before measurement.
- Linear vibrator should be mounted to the center.
- Also, acceleration sensor should be mounted the Dummy Jig's back corner.
- At measurement of acceleration, Dummy Jig should be stabilized.
- When use customer requirements JIG, according to customer conditions.

4. Characteristics

No.	Item	Specifications	Conditions & Remarks
4-1	Motor Diameter	Φ4±0.1 mm	See appendix 1
4-2	Motor Height	1.5±0.1 mm	See appendix 1
4-3	Weight	0.1±0.01 Gram	Motor（without line）
4-4	Resistance	13±2 Ω	25 ± 3 ℃
4-5	Rated Current	Max. 50 mArms	Input Source: F0, 0.4Vrms AC, sine wave
4-6	Vibration Acceleration	Min 0.12 Grms	Input Source: F0,0.4VrmsAC,sine wave, Use 10g JIG
4-7	Operating Frequency	240±10 Hz	See figure 2
4-8	Rising Time	Max. 40 msec	From 0 to 50% of nominal vibration @ 0.4Vrms, F0
4-9	Falling Time	Max. 60 msec	From 100% to 50% of nominal vibration @ 0.4Vrms, F0
4-10	Mechanical Noise	Max. 45 dB(A)	See figure 3 10cm distance from microphone, Input Source :F0,0.4Vrms AC,sine wave
4-11	Insulation Resistance	Min. 10 MΩ	100V DC input, between Case and Land of FPC.

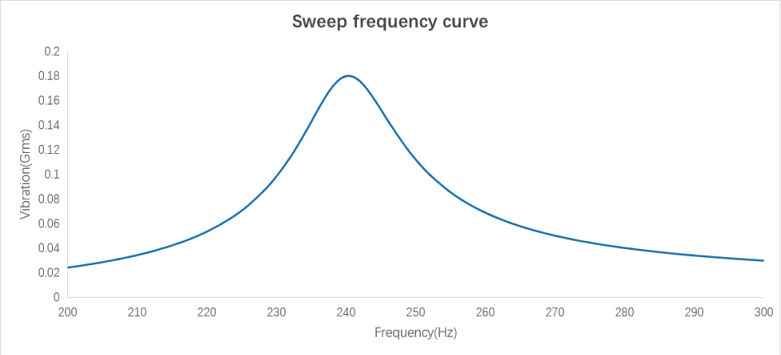


Figure2: Typical Acceleration characteristics

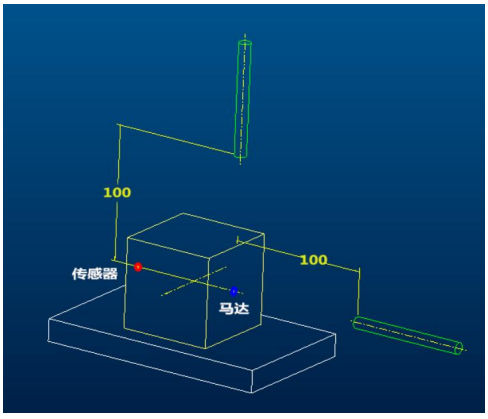
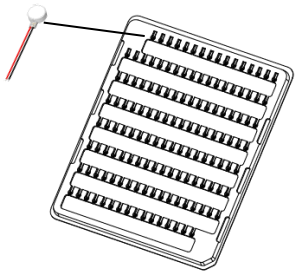


Figure3: Noise measurement condition

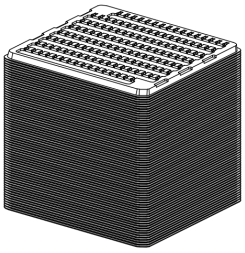
5. Reliability test

TBD

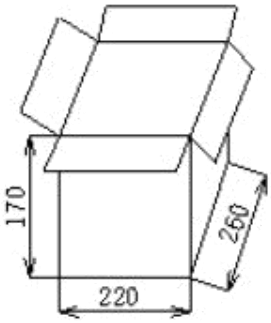
6. Packing



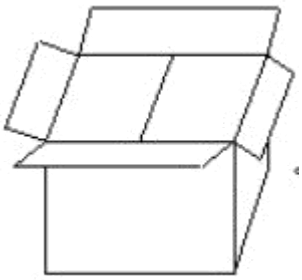
100pcs/plate
(about 70g)



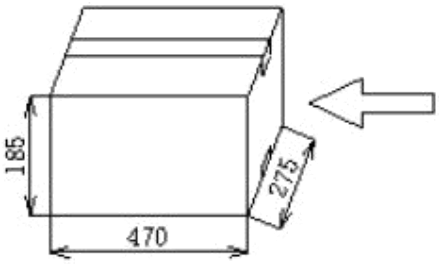
22 layers/bundle
(about 1520g)



1bundle/box
(about 1620g)



2 inner boxes/master carton



4000pcs/master carton
(about 3.5Kg)

7. Cautions & Handling

(1) Do not press the product with more than 0.5 kgf or drop it.
It can cause the transformation of performance or external appearance.

(2) Don't use under the following conditions. It may cause a decline in performance.
- Do not drop into fluid (such as: water, alcohol, etc.)
- Do not keep at high temperature or high humidity for extended periods of times
- Do not use near gases which cause erosion
- Please refrain from operating the vibrator near magnetic devices.

(3) The vibrator has a strong magnet, so please be aware that it has a magnetic force on the surface of the bracket.

(4) To optimize the vibration force, rated frequency and voltage could be changed as to assemble condition.

8. Appendix 1 (Outline drawing)

